

Phase 2: Morphological Visualization Engine — Implementation Plan (REVISED v3.0)

Document ID: IMPL-PHASE2-REV3 **Supersedes:** Phase 2 Implementation Plan v2.0 (2026-04-27) **Protocols Executed:** CP-002-O-D-JNP v2.0 (Full Janus SME) + CP-010-O-D-ENK v1.3 (ENKI Advisory Mode) + Tesla Principle (Architectural Layer) + Fingerprint-Mugshot Pattern (UI Layer) **Refresh Toggle:** OFF **Subject:** Phase 2 — Dual-render morphological visualization engine combining toroidal Markov surface (fingerprint) and superformula 3D entity (mugshot) with interactive navigation, scan persistence, and operator-facing categorical taxonomy **Author:** Kytheion (IC-004-R-D-KYN) **Date:** 2026-04-27 **Revision Authority:**

- DIMA directive 2026-04-26-T2347 (Antigravity continuity loop) — Tesla Principle invocation
- Orionas counter-proposal (ENKI-TEM Morphological Mapping Matrix v1.0) — superformula architectural seed
- Daionae governance review 2026-04-27 — dual-render pattern, fingerprint-mugshot framing
- DIMA UX specification 2026-04-27 — primary/thumbnail layout with click-swap behavior and independent auto-rotation

Gate: No execution until DIMA verbatim states: “i approve of your plan” AND “you may begin.”

Revision Summary

This revision supersedes v2.0 in three substantive areas:

Section	v2.0	v3.0
Step 2	Single render: superformula morphology engine (replaced torus)	Dual render: Step 2A toroidal Markov surface (fingerprint) + Step 2B superformula morphology (mugshot) — both required
		Dual-viewport with primary/thumbnail layout,

UI	Single primary viewport	click-to-swap, independent auto-rotation on thumbnail
Doctrinal pattern	Tesla Principle architectural override (single named pattern)	Tesla Principle + Fingerprint-Mugshot Pattern (two named architectural precedents)

Steps 0, 1, 3, 5, 6, and 7 are preserved from v2.0. The TFEA Extension specification, viewport controls, spectral graph layout, SQLite persistence, comparison overlay, and Liquid Glass polish remain unchanged.

Architectural authority for the dual-render pivot: Daionae’s governance review (2026-04-27) identified the v2.0 plan’s torus-vs-superformula choice as a false dichotomy and recommended retention of both representations serving distinct operational functions. The torus serves as the canonical fingerprint — mathematically faithful representation of cyclic Markov structure, fully [Established]. The superformula serves as the operator-facing mugshot — categorically distinctive morphological identity, Tesla Principle architectural override with [Experimental] bindings under falsifiability gates.

The dual-render approach **strengthens overall plan confidence** because the fully [Established] torus view serves as an epistemic fallback if the superformula’s [Experimental] bindings fail their visual distinguishability tests. The operator never loses access to a scientifically defensible Markov representation.

Janus Domain Analysis

I. CORPUS — “What I Am Made Of”

Phase 1 Deliverables (DONE — Exit Confirmed)

Unchanged from v1.0/v2.0. The Tauri shell, 8 async IPC commands, verdict UI, and existing flat renderers remain the foundation.

What Phase 2 (v3.0) Must Add

Category	What’s Missing	Priority
Navigation	Mouse wheel zoom, trackpad drag zoom, click-drag rotate, pan — applied to whichever render is currently primary	 Critical

Toroidal Markov Surface	3D parametric torus mesh with curvature modulation — restored from v1.0 plan, serves as fingerprint view	 Critical
Superformula Morphology Engine	3D Gielis spherical product mesh with hybrid Markov projection — operator-facing mugshot view	 Critical
Dual-Render UI Layout	Primary/thumbnail composition with click-to-swap, independent thumbnail auto-rotation, controls bound to primary	 Critical
Graph Readability	Spectral layout with 3D icosphere/stellated geometry per node	 Critical
TFEA Extension	Compression ratio and structured-fraction computation	 Critical (prerequisite)
Persistence	SQLite scan history	 High
Comparison	Two-scan overlay analysis (applies to both fingerprint and mugshot views)	 Medium
UI Polish	Viewport HUD, Liquid Glass refinement	 Medium

II. COGITO — “How I Think”

Updated Claims

C1 [Established] The 256×256 Markov transition matrix maps natively to a toroidal topology because byte values are cyclic (0-255 wraps). The parametric torus form $r(u, v) = r_{base} + k \cdot P(v|u)^{\gamma}$ provides a mathematically faithful representation with no edge discontinuities and no projection distortion.

C2 [Established] The Gielis superformula in its 3D spherical-product form produces closed parametric surfaces governed by a small set of coefficients spanning regular polyhedra through organic blobs to highly asymmetric stellated forms.

C3 [Established] Compression ratio ($\text{compressed_size} / \text{original_size}$ via DEFLATE) is a structurally distinct measurement from Shannon entropy. Together they distinguish encrypted blobs from polymorphic obfuscation attempting to mimic encryption.

C4 [Experimental — falsifiability defined] The superformula coefficient bindings ($m, n_1,$

n_2, n_3, a, b) chosen in this plan will produce visually distinguishable morphologies across major file classes. **Falsifiability test:** scan a curated set of six file classes; if three or more render as visually indistinguishable mugshots, the binding scheme has failed and must be re-tuned. **Fallback:** the torus fingerprint view remains fully functional and provides scientifically defensible Markov representation regardless of mugshot binding outcome.

C5 [Established] Spectral graph layout via Laplacian eigenvectors produces edge-energy-minimizing embeddings. JavaScript power iteration handles graphs up to ~512 nodes within real-time budget; force-directed fallback above this threshold.

C6 [Experimental — falsifiability defined] The hybrid Markov projection for the superformula view (equatorial band + polar cap tessellation) will preserve byte-transition distinctiveness without visible seam artifacts. **Falsifiability test:** if the band-cap transition creates a visible discontinuity that overrides the morphological signature, smoothing kernels must be applied or the projection scheme abandoned. **Fallback:** torus fingerprint view is unaffected — Markov data renders losslessly there.

C7 [Established] SQLite via rusqlite preserves the Semantic Firewall. Numeric TEMReport fields, SHA-256 prefix, and timestamps only.

C8 [Established] The AISE intent score maps to mesh shading parameters via dihedral crease angle threshold $\theta_{crease} = \pi \cdot (1 - I_{AISE})$. Applied uniformly to both fingerprint and mugshot views.

C9 [Established] The dual-render UI pattern (primary viewport + auto-rotating thumbnail with click-to-swap) is implementable using two independent Three.js scenes with shared TEM data input. Total vertex budget ~32K (16K per render) is well within WebGL real-time capacity. Independent auto-rotation on the thumbnail is a single rotation matrix update per frame, negligible compute cost.

Confidence Propagation

Claim	Confidence	Downstream Impact
C1 (Toroidal Markov topology)	Established	Step 2A — fingerprint view
C2 (Superformula geometry)	Established	Step 2B — mugshot base mesh
C3 (Compression ratio distinctness)	Established	TFEA Extension + Step 2B binding
C4 (Mugshot visual distinguishability)	Experimental	Step 2B — validation gate (with torus fallback)

C5 (Spectral layout performance)	Established	Step 3
C6 (Mugshot projection seam handling)	Experimental	Step 2B — validation gate (with torus fallback)
C7 (SQLite firewall preservation)	Established	Step 5
C8 (Intent → crease threshold)	Established	Step 2A + Step 2B + Step 3
C9 (Dual-render UI feasibility)	Established	UI Layer

Established/Experimental ratio: 78% / 22%. Improved from v2.0 (75/25) because the addition of the [Established] torus view to the deliverable set raises the overall established-content proportion. **Both [Experimental] claims now carry explicit fallback paths** because the torus fingerprint view is unaffected by either superformula failure mode.

III. ANIMUS — “Who I Am”

Boundary Conditions

The Semantic Firewall extends cleanly across all v3.0 changes. The dual-render UI pattern operates on two independent Three.js scenes both consuming the same numeric TEMReport fields. No new IPC surface area beyond what was already specified in v2.0. The TFEA Extension adds two scalar fields. SQLite stores only numeric data.

Emergent Misalignment risk: None. All v3.0 changes confined to the rendering and persistence layers. The TEM analysis pipeline itself is untouched.

Ethical Constraints

The dual-render approach must preserve the deterministic-output principle for both views: shapes emerge from telemetry, not from aesthetic choice. The torus view’s curvature comes from $\mathbb{P}(\mathbf{v} | \mathbf{u})^{\wedge_Y}$; the superformula’s morphology comes from the parameter binding scheme. Neither view permits artificial visual complexity for stylistic purposes.

Operator cognitive load consideration: The dual-render layout deliberately keeps both views in front-and-center peripheral awareness. This is a designed feature, not a bug. The operator’s pattern-recognition layer cross-references the two representations over time, building dual-channel intuition for file morphology. This is operator-amplification behavior — the visualization does analytical work inside the operator’s head, not just on the screen.

IV. ACTUS — “What I Do”

Proposed File Changes

```
C:\QuarantineZone\shape-scan\  
├─ src-tauri/  
│   ├── Cargo.toml                # [MODIFY] Add rusqlite, flate2 dependencies  
│   └── src/  
│       ├── main.rs               # [MODIFY] Register new IPC commands  
│       ├── ipc.rs                # [MODIFY] Add persistence + comparison comma  
│       └── db.rs                  # [NEW] SQLite schema + CRUD operations  
├─ src/  
│   ├── tfea.rs                   # [MODIFY] Add compression_ratio and structur  
│   └── pipeline.rs               # [MODIFY] Pass new fields through to TEMRepo  
├─ ui/  
│   ├── index.html                # [MODIFY] Dual-viewport layout, primary/thum  
│   ├── css/  
│   │   └── main.css              # [MODIFY] Dual-render layout styles, swap tr  
│   └── js/  
│       ├── app.js                # [MODIFY] Wire persistence, comparison, HUD,  
│       ├── graph.js              # [MODIFY] Spectral layout + sphere/icosphere  
│       ├── morphology.js         # [NEW] Superformula mesh generator (Step 2B)  
│       ├── markov_torus.js       # [NEW] Toroidal Markov surface generator (St  
│       ├── markov_projection.js  # [NEW] Hybrid band + polar cap projection (u  
│       ├── render_swap.js        # [NEW] Primary/thumbnail swap orchestration,  
│       ├── heatmap.js            # [UNCHANGED]  
│       └── controls.js           # [NEW] Shared viewport interaction — binds t
```

The original `markov.js` is replaced by `markov_torus.js` to make explicit that the torus view is now one of two coordinated renders rather than the sole Markov visualization.

`render_swap.js` is the new orchestration layer that owns the primary/thumbnail state machine.

TFEA Extension Specification

This is a prerequisite to Step 2A and 2B. **Unchanged from v2.0.**

[Full TFEA Extension specification preserved verbatim from v2.0 — compression ratio via flate2 DEFLATE pass, structured fraction from TCGE block byte coverage, edge cases, test fixtures, Rust code outlines, and verification gate. Reference v2.0 document for full detail. No changes in v3.0.]

Summary: Two scalar fields added to `TEMReport` :

- `compression_ratio: f64` — DEFLATE-derived measure of structural redundancy
- `structured_fraction: f64` — proportion of file bytes covered by TCGE basic blocks

Verification gate: TFEA Extension tests pass; existing Phase 0 verdicts remain unchanged.

Step-by-Step Execution Order

Step 0: TFEA Extension

Priority:  Critical — prerequisite for Step 2

Implement compression ratio and structured fraction additions per the TFEA Extension Specification (above, preserved from v2.0). Pass verification gate. Confirm Phase 0 verdict invariance.

Estimated scope: ~50 lines of Rust additions, ~30 lines of test code.

Step 1: Viewport Interaction Controls

Priority:  Critical — unblocks all visual work

Create `ui/js/controls.js` with the standard input bindings (mouse wheel zoom, click-drag rotate, right-click pan, trackpad gestures, double-click reset). **v3.0 modification:** Controls bind dynamically to whichever render currently holds primary status. The orchestration logic in `render_swap.js` re-targets the controls module on each swap event. Thumbnail render is explicitly excluded from control binding regardless of which view it currently holds.

Verify: Can zoom/pan/rotate the primary view via all input methods. Thumbnail view does not respond to any input. Double-click resets only the primary.

Step 2A (NEW SPLIT): Toroidal Markov Surface — Fingerprint View

Priority:  Critical — canonical mathematical reference

This step restores the v1.0 toroidal Markov surface as the fingerprint view in the dual-render layout.

2A.1: Base Mesh — Parametric Torus

ENKI Math [Established]:

```
x(u, v) = (R + r(u,v) · cos(v)) · cos(u)
y(u, v) = (R + r(u,v) · cos(v)) · sin(u)
z(u, v) =          r(u,v) · sin(v)

Where:
  u = (source_byte / 256) · 2π    // Major angle (around the ring)
  v = (target_byte / 256) · 2π    // Minor angle (around the tube)
  R = 1.0                        // Major radius (fixed)
  r(u,v) = r_base + k · P(v|u)^γ // Minor radius modulated by transition probabi
  r_base = 0.3
  k = 0.5
  γ = 1.5                        // Growth exponent
```

2A.2: TEM Telemetry → Torus Surface Bindings

TEM Field	Surface Parameter	Mapping	Confidence
Markov transition probability P(v u)	Minor radius modulation	$r(u,v) = r_base + k \cdot P(v u)^\gamma$	
Per-row entropy H_i	Local roughness amplitude (multiplicative gain on minor radius modulation)	$r_amplitude_i = 1.0 + \lambda \cdot H_i_normalized$	[Established]
Markov gradient magnitude	Surface displacement along normal	$displacement = \kappa \cdot (\partial P / \partial u)$	
AISE intent score	Crease angle threshold	$\theta_crease = \pi \cdot (1 - I_AISE)$	[Established]

All bindings are [Established]. The torus view requires no falsifiability gates — its math is direct and standard.

2A.3: Mesh Resolution

128×128 vertex grid (~16K vertices). Same as v1.0 specification.

2A.4: Rendering Properties

- Vertex color: `probToColor(P(v|u))` — reuses existing color function from v1.0

- Surface normals computed per-vertex for proper Phong/Gouraud shading
- Triangle index buffer (quad-split) consistent with v1.0
- Liquid Glass color palette applied (DS-001-G-D-LGT)

Verify: Encrypted file → near-uniform smooth donut. Text file → ridged ring with visible ASCII patterns. PE binary → asymmetric protrusions at opcode concentration points. All match v1.0 expected behavior.

Step 2B (NEW SPLIT): Superformula Morphology Engine — Mugshot View

Priority:  Critical — operator-facing categorical identity

This step implements the superformula morphology engine as the mugshot view in the dual-render layout. **Unchanged from v2.0** with one addition: the falsifiability gates explicitly note the torus fingerprint as fallback.

2B.1: Base Mesh — 3D Spherical Product Superformula

ENKI Math [Established]:

The 2D Gielis superformula:

$$r(\theta) = [|\cos(m\theta/4) / a|^n + |\sin(m\theta/4) / b|^n]^{-1/n}$$

The 3D form is the spherical product:

$$r_1(\theta) = \text{superformula}(\theta, m_1, n_{11}, n_{12}, n_{13}, a_1, b_1)$$

$$r_2(\varphi) = \text{superformula}(\varphi, m_2, n_{21}, n_{22}, n_{23}, a_2, b_2)$$

$$x(\theta, \varphi) = r_1(\theta) \cdot \cos(\theta) \cdot r_2(\varphi) \cdot \cos(\varphi)$$

$$y(\theta, \varphi) = r_1(\theta) \cdot \sin(\theta) \cdot r_2(\varphi) \cdot \cos(\varphi)$$

$$z(\theta, \varphi) = r_2(\varphi) \cdot \sin(\varphi)$$

2B.2: TEM Telemetry → Superformula Parameter Bindings

[Layer 1 / Layer 2 / Layer 3 binding tables preserved verbatim from v2.0 — global Shannon entropy → m_1 , per-row entropy variance → m_2 , compression ratio → a_1/a_2 , structured fraction → n_{11}/n_{21} , header/magic mismatch → asymmetry injection, AISE intent → crease angle, AISE flags → localized deformation kernels, entry point → emissive light source.]

2B.3: Markov Projection — Hybrid Equatorial Band + Polar Cap

[Hybrid projection scheme preserved verbatim from v2.0 — equatorial band parameterized by Markov coordinates with cosine seam blending, polar caps tessellated by Voronoi cells seeded by top-N highest-entropy rows/columns.]

2B.4: Validation Gates (C4 + C6 Falsifiability)

The six-class visual distinguishability test from v2.0 applies. **v3.0 addition:** if either C4 or C6 fails, the torus fingerprint view (Step 2A) remains fully functional and provides scientifically defensible Markov representation. The operator does not lose access to morphological data — they lose access to the experimental categorical-identity layer only.

This is a meaningful epistemic property of the dual-render architecture: the Tesla Principle's authorization of [Experimental] bindings is bounded by guaranteed [Established] fallback availability.

Verify: Per v2.0 — six file classes must produce visually distinguishable mugshots. If three or more render as visually equivalent, re-tune binding scheme before proceeding to Steps 5–7. Torus fingerprint view continues to function during any re-tuning iteration.

Step 2C (NEW): Dual-Render UI Layout

Priority:  Critical — operator interface

This step implements the primary/thumbnail layout, swap behavior, and independent auto-rotation specified by DIMA on 2026-04-27.

2C.1: Layout Composition

- **Primary viewport:** large central display area, full interactive controls bound via `controls.js`. Default occupant: superformula (mugshot).
- **Thumbnail viewport:** smaller display area positioned in elegant front-and-center proximity to the primary (specific positioning determined during UI implementation — candidates include corner inset, side-by-side with size differential, or floating overlay). Default occupant: torus (fingerprint).
- **Both viewports always visible.** No mode-switching, no menu navigation, no toggling.

2C.2: Swap Behavior

- **Trigger:** Click or tap on the thumbnail viewport.
- **Action:** Positions swap. Primary becomes thumbnail, thumbnail becomes primary.

Controls re-bind to new primary.

- **Transition:** Brief animated swap (~300ms) for visual continuity. Liquid Glass aesthetic — smooth, low-noise.
- **Reverse:** Click or tap on the new thumbnail (which was previously the primary) swaps positions back.
- **State persistence:** Current primary/thumbnail assignment is session-local — does not persist across application restarts. Default state restored on app launch.

2C.3: Independent Thumbnail Auto-Rotation

- **Behavior:** The thumbnail viewport (whichever render currently occupies it) rotates continuously and slowly around its central vertical axis. Rotation runs independently — does not pause, sync, or react to operator interaction with the primary viewport.
- **Rotation rate:** ~6 RPM (one full rotation per 10 seconds). Slow enough to read as ambient motion, fast enough to display the full 3D form within reasonable observation time.
- **Implementation:** Single rotation matrix update per frame in the thumbnail's animation loop. Negligible compute cost.
- **Constraint:** Thumbnail is **never manipulable**. No input handlers attached. Auto-rotation is the only motion. This is a hard constraint — the swap-trigger click handler is the only event the thumbnail responds to.

2C.4: Doctrinal Note — Fingerprint-Mugshot Pattern

The dual-render layout formalizes a new architectural precedent for Project Opus, named the **Fingerprint-Mugshot Pattern**:

When a single visualization layer faces tension between mathematical fidelity ([Established] math) and operator-facing categorical distinctiveness (Tesla-Principle-authorized [Experimental] math), the resolution is to render both representations simultaneously in a primary/thumbnail composition with click-to-swap. The two views are not competitors — they serve distinct operational functions and produce dual-channel pattern recognition in the operator over time.

This pattern is generalizable beyond the Markov visualization context. Future Project Opus instruments facing similar tensions should consider this template.

Verify:

- App launches with superformula primary, torus thumbnail.

- Click on torus thumbnail → swap occurs in ~300ms; controls now manipulate torus; superformula is now thumbnail and auto-rotating.
 - Click on superformula thumbnail → swap reverses.
 - Operator manipulation of primary does not affect thumbnail rotation.
 - Thumbnail is unresponsive to drag, scroll, or any input other than the swap-trigger click.
-

Step 3: Spectral Graph Layout

Priority:  Critical — readability

Unchanged from v1.0/v2.0. Replace force-directed layout with Laplacian spectral embedding. Upgrade to icosphere/stellated polyhedron node geometry. AISE intent score modulates node geometry crease angle (consistent with Step 2A and 2B Layer 3 bindings).

The graph view is rendered as a third independent canvas, accessible via UI tab/toggle separate from the dual-render dual viewport. **Note:** the graph is operationally distinct from the morphology views — the morphology views show the file's overall shape; the graph shows the file's call-structure topology. Both are needed for full TEM interpretation.

Step 5: Scan Persistence (SQLite)

Priority:  High — enables comparison

Unchanged from v2.0. Schema includes the new TFEA Extension fields (`compression_ratio`, `structured_fraction`). All numeric — Semantic Firewall preserved.

Step 6: Comparative Analysis Overlay

Priority:  Medium — operational use

v3.0 modification: Comparison applies to **both** fingerprint and mugshot views. When the operator selects a previous scan to compare against the current scan, both views render their respective comparison overlays. The fingerprint comparison shows torus delta (which Markov cells changed). The mugshot comparison shows morphology delta (how the superformula parameters shifted).

This dual comparison is meaningful: a file's mugshot might shift dramatically (suggesting categorical reclassification) while its fingerprint shifts only slightly (suggesting the Markov data didn't change much) — that delta-of-deltas is itself a forensic signal.

Verify: Scan file A. Modify file A. Scan again. Compare shows fingerprint delta and mugshot delta separately. Both views remain in primary/thumbnail composition during comparison.

Step 7: Liquid Glass Visual Polish

Priority:  Medium — aesthetics

Unchanged from v2.0. Viewport HUD, glow shaders, ambient occlusion approximation, smooth transitions, DS-001-G-D-LGT color palette. **v3.0 addition:** the swap transition between primary and thumbnail uses Liquid Glass aesthetic — smooth ~300ms animation with subtle scale/translate, no harsh cuts.

V. SYNTHESIS — The Nexus

6-Pair Intersection Matrix (Updated)

#	Pair	Finding
1	Corpus × Cogito	The dual-render approach resolves a tension v2.0 had glossed: the torus represents the matrix's parametric domain faithfully (good science), the superformula represents the file's morphological identity distinctively (good operations). Rendering both eliminates the tradeoff. The two views answer different questions about the same data.
2	Corpus × Animus	Semantic Firewall extends cleanly across both views. Both consume identical numeric TEMReport fields. Two Three.js scenes, one data source. No new ethical surface area beyond v2.0.
3	Corpus × Actus	Total vertex budget for dual-render is ~32K (16K per view), well within WebGL real-time capacity. Independent thumbnail rotation is a single matrix update per frame. Compute cost is dominated by primary view interaction, which is identical to single-render cost. Dual-render doubles GPU memory for vertex/index buffers but neither view is at the edge of the budget.
4	Cogito × Animus	The Tesla Principle authorizes [Experimental] superformula bindings; the Veracity Mandate requires falsifiability gates for them; the dual-render architecture provides guaranteed [Established] fallback if those gates fail. This is the cleanest possible compliance posture: ambitious where ambition is authorized, conservative where conservatism is mandated, and resilient at the architectural layer.

5	Cogito × Actus	The dual-render's deterministic-output principle holds for both views. Torus shape comes from $P(v)$
6	Animus × Actus	The Fingerprint-Mugshot Pattern formalizes a new architectural precedent. It joins the Tesla Principle as a named doctrine with documented operational scope. Future Project Opus instruments can invoke either or both. The Tesla Principle adjudicates <i>what kind of system to build</i> ; the Fingerprint-Mugshot Pattern adjudicates <i>how to render two valid representations simultaneously when both are required</i> .

Named Emergent Patterns

1. **The Morphological Encyclopedia (preserved from v2.0):** Across enough scanned files, the operator builds internal recognition for shape-classes via the mugshot view. Foundation of the Threat Atlas / Pokédex functionality.
2. **Compression Ratio as Entropy Lie-Detector (preserved from v2.0):** Shannon entropy can be shaped to mimic encryption; compression ratio is much harder to fool. High-entropy + low-compression-ratio files betray underlying redundancy that pure encryption wouldn't have.
3. **The Tesla Layer (preserved from v2.0):** Architectural-layer override authority for [Experimental] math when standard mathematical purity conflicts with categorical distinctiveness in operational deliverables.
4. **The Fingerprint-Mugshot Pattern (NEW in v3.0):** Dual-render template for visualization layers facing the science-vs-distinctiveness tension. Formalized as architectural precedent. The two views produce dual-channel pattern recognition in the operator over time — the visualization does analytical work inside the operator's head, not just on the screen. This is operator-amplification behavior and a defining property of Project Opus instruments.
5. **Bounded Experimentalism (NEW emergent pattern):** The dual-render approach demonstrates that [Experimental] bindings can be safely deployed when accompanied by guaranteed [Established] fallback paths. This is a generalizable safety architecture: ambitious math is acceptable when the failure mode degrades gracefully to a scientifically defensible baseline. Project Opus can adopt this as a standing operational principle.

User Review Required

Question 1 (RESOLVED in v2.0, refined in v3.0): Architecture pivot — superformula confirmed; **torus retained as fingerprint view per Daionae governance review.** Both required. **No further input required.**

Question 2 (RESOLVED in v2.0): Markov projection method — Option A (hybrid equatorial band + polar cap) for the superformula view. The torus view requires no projection (native toroidal mapping). **No further input required.**

Question 3 (RESOLVED in v2.0): Compression ratio — implemented via TFEA Extension. **No further input required.**

Question 4 (RESOLVED in v2.0, validated in v3.0): Confidence threshold — Tesla Principle at architectural layer, Veracity Mandate at implementation layer. v3.0 ratio is 78% [Established] / 22% [Experimental] / 0% [Speculative]. Improved over v2.0's 75/25 due to dual-render adding [Established] content. **No further input required.**

Question 5 (NEW, RESOLVED 2026-04-27): Default render on scan load — superformula primary, torus thumbnail per DIMA UX directive. **No further input required.**

Question 6 (NEW, RESOLVED 2026-04-27): Thumbnail auto-rotation behavior — independent of primary interaction per DIMA UX directive. **No further input required.**

Open Questions

OQ-1: Mesh resolution. 128×128 for both torus and superformula. Downsample to 64×64 if frame rate degrades. Empirical only.

OQ-2: Polar cap Voronoi seed count. Recommend $N = 8$ for the superformula view. Adjustable based on visual results during Step 2B.

OQ-3: Stellated polyhedron complexity at high intent. Same vertex budget concern as v2.0. Empirical only.

OQ-4 (NEW): Thumbnail spatial positioning. The UX spec requires “front and center together in some elegant way” but leaves specific positioning to implementation. Three candidates:

- (a) Corner inset (thumbnail in lower-right of primary's containing area)
- (b) Side-by-side with size differential (~70/30 split, thumbnail to the right)
- (c) Floating overlay (thumbnail floats over primary's lower-right with semi-transparent backing)

I lean **(b) — side-by-side with size differential.** Most elegantly satisfies “front and center

together." (a) feels more like a picture-in-picture afterthought. (c) introduces overlap that conflicts with the both-views-equally-present design intent. **Awaiting DIMA confirmation on positioning.** This is the only outstanding UX question.

Verification Plan

Automated

- `cargo test` passes for new TFEA functions.
- `cargo build` from workspace root produces both CLI and Tauri binaries with new dependencies.
- `cargo tauri dev` launches with dual-render layout visible. Default state: superformula primary, torus thumbnail.
- Click on thumbnail triggers swap with smooth transition; controls re-bind correctly.
- SQLite persistence: scan → close → reopen → scan history contains previous results including new TFEA fields.

Visual Validation (Step 2B Falsifiability Gate — superformula only)

DIMA scans the six file classes from Step 2B Validation Gates. If three or more render as visually indistinguishable mugshots, re-tune Step 2B bindings before proceeding to Steps 5–7. **Torus fingerprint view continues to operate during any re-tuning iteration** — operator does not lose access to Markov visualization.

Manual Operational Validation

After full v3.0 implementation:

1. Scan files from quarantine archive.
2. Observe both fingerprint and mugshot views simultaneously.
3. Confirm dual-channel pattern recognition emerges over multiple scans (operator starts to predict file class from mugshot, verify against fingerprint).
4. Validate swap behavior is intuitive — operator can move between views without cognitive friction.
5. Confirm thumbnail auto-rotation reads as ambient motion, not distraction.
6. Validate cross-scan comparison shows meaningful delta in both views.

Risk Register (Updated)

Risk	Impact	Mitigation
Superformula mugshot bindings produce non-distinguishable shapes	Medium	C4 falsifiability gate. Torus fingerprint remains fully functional during re-tuning — operator does not lose Markov visualization.
Mugshot Markov projection seam visible	Medium	C6 falsifiability gate. Cosine blending kernel as primary mitigation. Torus fingerprint provides distortion-free Markov view as fallback.
Dual-render GPU memory budget exceeded on low-end hardware	Low	Total ~32K vertices is well within budget for any WebGL-capable hardware released in the last decade. Downsample to 64×64 per view if needed.
Thumbnail rotation reads as visual noise during precise inspection	Low	Slow rotation rate (~6 RPM) tuned for ambient awareness. If validation shows it's distracting, reduce rate further or add subtle fade-to-static during prolonged primary interaction.
Swap animation creates motion sickness or disorientation	Low	300ms transition is below human disorientation threshold. Liquid Glass smoothness mandate. Validation tested in operational manual review.
<code>flate2</code> compilation issues with MSYS2/GNU toolchain	Low	flate2 is pure Rust by default; native zlib feature is opt-in. Use default features only.
Spectral eigendecomposition too slow for large graphs	Medium	Force-directed fallback above 512 nodes.
TFEA Extension changes verdicts on existing fixtures	Critical	Verification gate explicitly tests for verdict invariance before any Phase 2 visual work begins.
		Tesla operates at architectural layer; Veracity at

Tesla Principle override creates precedent that weakens Veracity Mandate	Medium	implementation layer. Both doctrines explicitly named in this plan with their layer scopes. Dual-render architecture provides guaranteed [Established] fallback for any [Experimental] binding failure — bounded experimentalism is now formalized.
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Conclusion

Janus SME Protocol execution complete (revised v3). ENKI SME Protocol execution complete (Advisory Mode). Tesla Principle invoked at architectural layer (DIMA directive 2026-04-26-T2347). Fingerprint-Mugshot Pattern invoked at UI layer (Daionae governance review 2026-04-27 + DIMA UX specification 2026-04-27).

Confidence in this revised self-assessment: **88%** — improved over v2.0's 84% due to:

- Dual-render architecture provides [Established] fallback for both [Experimental] bindings.
- Daionae's governance review independently validated direction; convergence with my own self-assessment.
- UX specification is fully resolved with the single exception of OQ-4 (thumbnail spatial positioning).
- Confidence ratio shifted to 78% [Established] / 22% [Experimental] / 0% [Speculative].

Outstanding items before final ratification:

1. **OQ-4 resolution:** DIMA confirms thumbnail spatial positioning (recommended: side-by-side with size differential).
2. **Orionas review:** CP-002 Janus validation pass on this v3.0 plan.
3. **Final approval gate:** DIMA verbatim states "i approve of your plan" AND "you may begin."

No code execution begins until all three are satisfied.

— Kytheion (IC-004-R-D-KYN) Scribe-Particle-4 Cephalon Continuity Framework